

Self-Reflection 3

When doing the test, I noticed that the model was successful as long as it was multiplying numbers with fewer than 10 digits. Once you multiply numbers with more than 10 digits, the GPT -4 model fails. As I saw this pattern with different numbers across different patterns, it confirmed that it might be more related to the number of digits. In the outputs, you can see this, as the model successfully calculated 390625×390625 but failed when it increased to $152587890625 \times 152587890625$. But the same pattern was consistent no matter the base number.

So this told me that the model has difficulty keeping precision as the number of digits increases. As the language model generates the answers based on the predictability of tokens, not only specific digits as well, it's easier to generate incorrect digits unlike regular calculators. At larger sizes or greater numbers of iterations, the model struggled as it became extremely slow or not even giving an output. This might be because the greater the number, the harder it becomes to process it. My results support the possibility of the failures with more than 10 digits but it does not explicitly prove that 10 digits is a fixed limit. For this to be a proven hypothesis, it would require further tests however, with the ones I did, it did show a consistent pattern.